WaRF with Jev: the Log-Odds Panel

Technical Note

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1 Summary

WaRF was run with TypeSafe's Jev as its only model, on real defects from the G+Smo history. A panel of five persona seats, three on `jev-1.14` and two on `jev-1.13`, aggregated in WaRF's log-odds mode after learning from 132 labelled reviews (LO), reviewed 220 fresh changes correctly in **73.2%** of cases (balanced accuracy), against **67.7%** for one Jev. This is the first preregistered test in which WaRF beat its single reviewer (exact McNemar $p = 0.012$, Holm-adjusted 0.024). Because it was the sixth comparison on those changes, LO was then frozen and tested once on a second project, MFEM: on 300 fresh changes it scored **80.0%** against **75.3%** for one Jev (23 against 9 changes, $p = 0.020$), with a calibration learned only on G+Smo. One Claude Opus scores 85.5% on the G+Smo fresh set.

2 Data and Measure

Defective changes C++ changes in the gismo6 history that a later fix blamed (B-SZZ), kept only if two independent judges, who see the fix, agree that the change introduced the defect. The reviewers never see the fix.

Clean changes Changes whose added lines all survive unchanged at `origin/dev`, one per defective change, matched by date and size.

Development set 132 changes (66 + 66), used for all tuning and learning.

Fresh set 220 changes (110 + 110), never used for tuning; every test on it was written down before the first call.

Input The diff with 30 lines of context and the enclosing function bodies, as WaRF sends it.

Measure Balanced accuracy (BA), the mean of the share of defects caught and the share of clean changes passed. Counting added lines alone ranks the fresh set with AUC 0.79, so size is a strong confound.

3 Jev

Jev (TypeSafe System One) is a decision model: it takes a state and typed questions and returns, per question, a probability for each option, with no text. WaRF asks one choice question, APPROVE or REJECT, with the seat's persona as its instructions. Two versions are served through OpenRouter: `jev-1.14` (2026-10-14) and `jev-1.13` (2026-09-17). A call costs about \$0.0001–0.0002; an LO review (five calls) about \$0.0006.

Jev is stable but forms one opinion per change: identical calls agree on the decision in 97–99% of cases, and its scores correlate 0.84–0.97 across wordings, input formats, retrieved context and the two versions.

4 The Log-Odds Panel

4.1 Seats

Seat	Persona	Model
SKEPTIC	defensive, concrete defects only	`jev-1.14`
ADVOCATE	author's intent	`jev-1.14`
AUDITOR	neutral audit	`jev-1.14`
TEST-FOCUSED	coverage	`jev-1.13`
BOUNDARY	edge cases	`jev-1.13`

4.2 Aggregation (ADR 0001)

Each seat s returns $p_s = p(\text{REJECT})$, read to two decimals and clamped to $[0.01, 0.99]$. Its log-odds for APPROVE is $r_s = \ln\big((1 - p_s)/p_s\big)$.

Calibration. From the labelled history of that seat (same persona, model and evidence source), WaRF fits Platt scaling once it has at least 10 labelled reviews of each class:

$$e_s = a_s + b_s\, r_s - \pi_s,$$

where π_s is the logit of the smoothed clean share of the training labels, so e_s is a likelihood ratio rather than a posterior. Without enough history, r_s is capped at the seat's vote-mode step $\omega_s \cdot \text{LLR}$.

Pooling. Seats of one model shown the same input form one group g; a group contributes the mean of its seats' evidence, and the groups are summed:

$$\Lambda = \sum_g \frac{1}{|g|} \sum_{s \in g} e_s.$$

The three `jev-1.14` seats thus count as one source and the two `jev-1.13` seats as another (SPECIFICATION §3.1: one model is one fault domain sampled n times).

Decision. After each complete group, WaRF stops if $\Lambda \geq A \approx 2.890$ (APPROVE) or $\Lambda \leq B \approx -2.251$ (REJECT), unless two groups point in opposite directions by at least $\ln 2$ each. If no boundary is crossed, it decides APPROVE if $\Lambda \geq \tau$, with $\tau = \operatorname{logit}\big((1+\theta)/2\big)$: 0 for LOGIC, 0.405 for SECURITY, -0.201 for PERFORMANCE.

4.3 Learning

The panel reviewed the 132 development changes; after each review the known answer was recorded with `apply_oracle` and `append_session`, WaRF's own labelling path. The resulting session log is LO's calibration history. No fresh change entered it.

5 Results on the Fresh Set

Reviewer (220 fresh changes)	Right	Caught	False alarms
One Jev (usual question, threshold from dev)	67.7%	73/110	34/110
WaRF Jev, 7 narrow checks per hunk	71.4%	67/110	20/110
WaRF Jev, narrow checks of both versions	71.8%	70/110	22/110
WaRF panel, untrained, vote mode	69.5%	85/110	42/110
WaRF panel, learned weights, vote mode	69.1%	85/110	43/110
LO: WaRF panel, learned, log-odds mode	**73.2%**	83/110	32/110
One Claude Opus, default effort	85.5%	91/110	13/110
WaRF Claude, three efforts, reject if any rejects	86.8%	100/110	19/110

LO against one Jev: only LO right on 16 changes, only one Jev on 4; $p = 0.012$, Holm 0.024 over the two arms of that test. The preregistered verdict is a win. LO against the untrained panel: +3.6 points ($p = 0.077$); against the narrow checks: +1.8 points ($p = 0.61$).

The seats' answers are the same in the vote-mode and log-odds arms; the gain comes from the aggregation. Calibrated evidence pooled per model keeps most of the extra defects the panel catches while cutting its false alarms from 42 to 32.

6 Second Repository: MFEM

Because the G+Smo fresh set had been used for six comparisons, LO was frozen and tested once on a second C++ project, MFEM (github.com/mfem/mfem, read with git only). The test was preregistered before any change was built.

Set B-SZZ candidates from 2,949 fix commits on `master`; judge A judged 360 candidates (157 yes), judge B judged those 157 (151 yes); the first 150 confirmed defects were kept, with 150 surviving clean changes matched by date and size.

LO Exactly as tested on G+Smo, calibrated only on the 132 G+Smo development reviews; nothing from MFEM entered it.

One Jev The usual question, REJECT at $p \geq 0.79$ (fixed on G+Smo).

Reviewer (300 MFEM changes)	Right	Caught	False alarms
One Jev	75.3%	94/150	18/150
WaRF Jev, 7 narrow checks per hunk	74.3%	88/150	15/150
LO: WaRF log-odds panel	**80.0%**	112/150	22/150

LO against one Jev: only LO right on 23 changes, only one Jev on 9; exact McNemar $p = 0.020$. The preregistered verdict is a win. LO also beats the narrow checks (29 against 12, $p = 0.012$); the narrow checks do not beat one Jev here. On the 87 defects visible from the diff LO scores 85.2% against 80.8%. Ranked by the seats' mean p, LO's AUC is 0.883 against 0.866 for the count of added lines: the gain lies in WaRF's decision, not in a better ranking.

7 Development of LO

LO's decision depends only on the seats' stored p_s and the labelled history, so it was rebuilt offline with WaRF's own calibrator, grouping and decision functions. The rebuild reproduces all 220 logged decisions. Variants were judged by stratified 10-fold cross-validation, repeated five times, over all 352 judged changes, calibrating on the training folds only; no further look at the fresh set was spent. Repeats vary by about ±0.4 points.

Cross-validated, 352 changes	Right
One Jev seat in WaRF (usual question, calibrated)	73.8%
LO, five personas	**76.3%**
LO + the usual question on both versions	76.7%
LO + the 7 narrow checks of both versions	76.4%
LO + usual question + narrow checks	75.5%
LO + a context seat (retrieved declarations)	75.2%

More Jev seats do not move LO: they repeat what the personas already say. A context seat, shown the declarations of the names on the changed lines, helps a little on its own (74.1%) but lowers LO: as a separate source its evidence is summed with the two Jev groups although it mostly repeats them.

7.1 Cold Start

A new user has no labelled history, so every seat starts uncalibrated:

Labelled reviews before the fresh set	LO right
none	68.6%
10 defective + 10 clean	70.5%
15 + 15	71.4%
20 + 20	73.6%
66 + 66 (development set)	73.2%

Without history LO is about one Jev. It needs 10–20 confirmed defects, which in a real project may take months to collect; clean labels are easy.

8 Limitations

- Both projects are C++ numerics libraries (isogeometric analysis and finite elements); other kinds of code are untested.

- Jev's ranking stays close to a count of added lines (G+Smo fresh set 0.75–0.77 against 0.79; MFEM 0.88 against 0.87); much of any Jev score is change size.
- LO asks five seats where one Jev asks one: about $0.00065 against $0.00013 per review.
- The labels rest on two Claude judges who see the later fix.
- The clean controls are clean only in that their lines survived; a few may hold unfound defects.
- Claude Opus remains about 12 points ahead of LO.

9 Files

- `.warf-experiments/confirm-set/README.md`: the preregistered tests and their results (second to fifth test).
- `.warf-experiments/confirm-set/jev-learned/`: LO's learning log, profiles, runner and scorer.
- `.warf-experiments/mfem-set/`: the MFEM preregistration, mining, judgements, items, runs and score.
- `.warf-experiments/lo-dev/`: the offline rebuild, cross-validation, learning curve and context seat.
- `.warf-experiments/real-set/DEV.md`: everything tried on the development set.